Analyze the situation, select an appropriate searching/sorting technique, justify the choice, implement it in c++, evaluate efficiency (Worst case, average case and best case), Solve one example showing all the steps to get the output by considering the suitable input data and write the test cases considering pass as well as fail status.

Sample Problem Statement:

1. **Hospital Record Processing** : A hospital has separate departments generating patient records. Each department independently sorts its patient IDs. At the end of the day, the hospital needs one globally sorted list.

   For example:

   OPD: 101, 105, 112, 120
   Emergency: 103, 108, 115, 121
   ICU: 102, 109, 116, 125

Performance Criteria for evaluation :

1. Realization of problem statement – 5M
2. Optimal / efficient implementation -5M
3. Documentation (**Solve one example showing all the steps to get the output by considering the suitable input data and write the test cases considering pass as well as fail status. Evaluate efficiency (Worst case, average case and best case)) -5M**
4. Performance in Question and answer session -5M